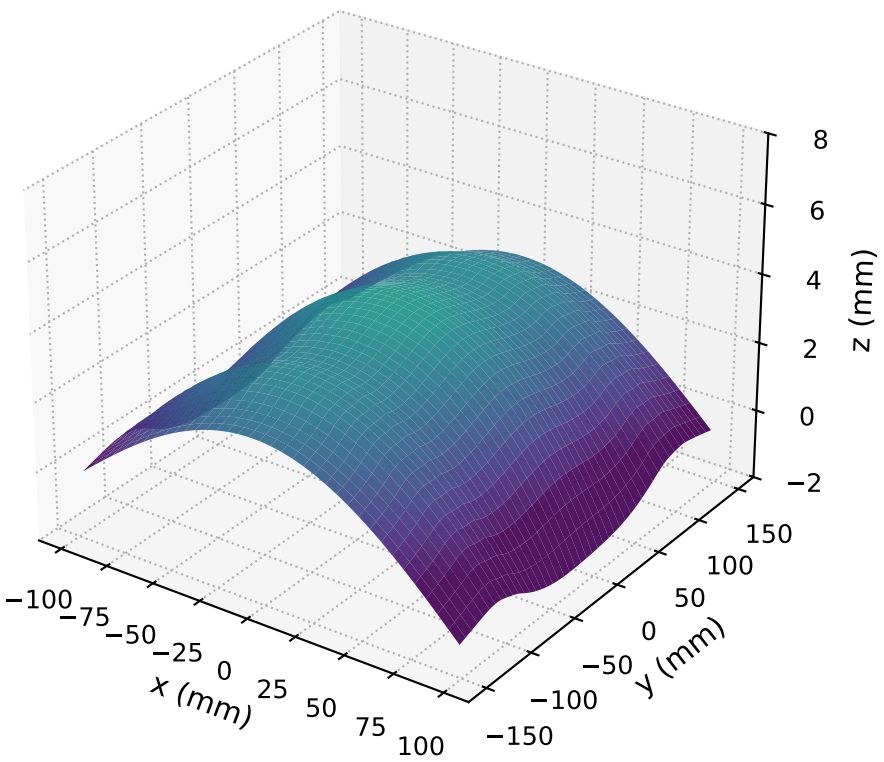
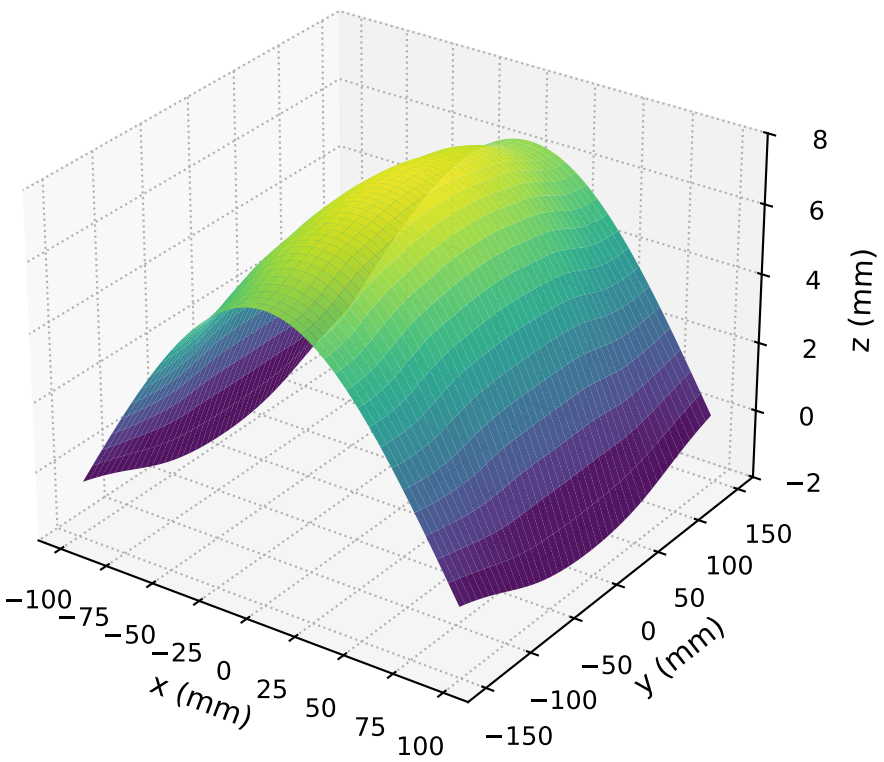
Row 1: Deformed 3D Equilibrium Surface (Elevation z [mm])

(a) Attempt 045 Baseline (Peak z = 4.03 mm)
Shallow dome with prominent boundary flaring



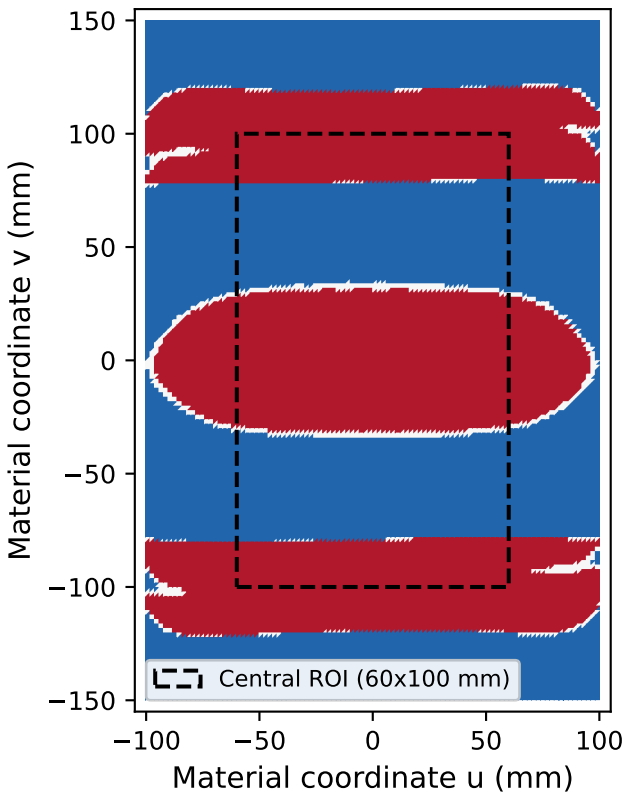
(b) Proposed Concentric Crosshatch (Peak z = 7.50 mm)
High-amplitude synclastic dome (+86% height)



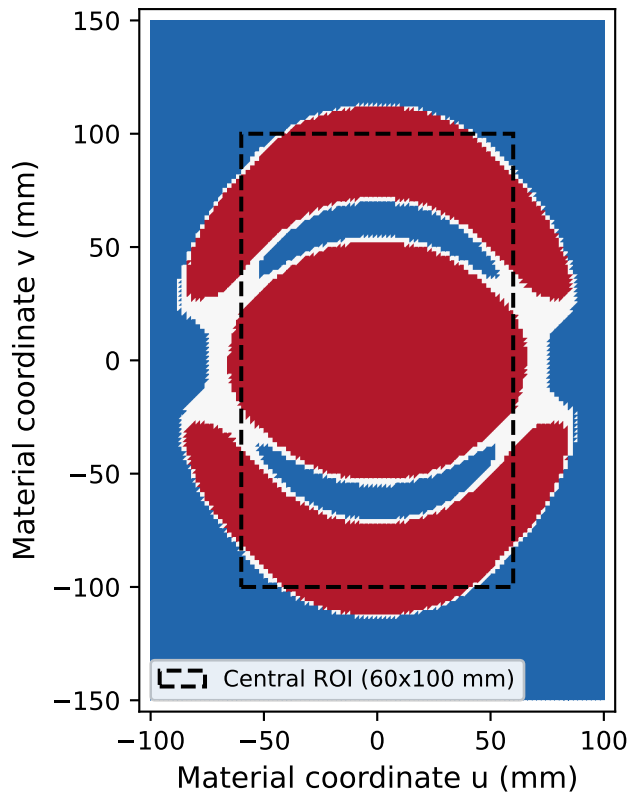


Row 2: Gaussian Curvature Sign Classification (Material UV Space)

(c) Attempt 045: 53.9% Saddle Flux
Severe blue saddle bands dividing sheet

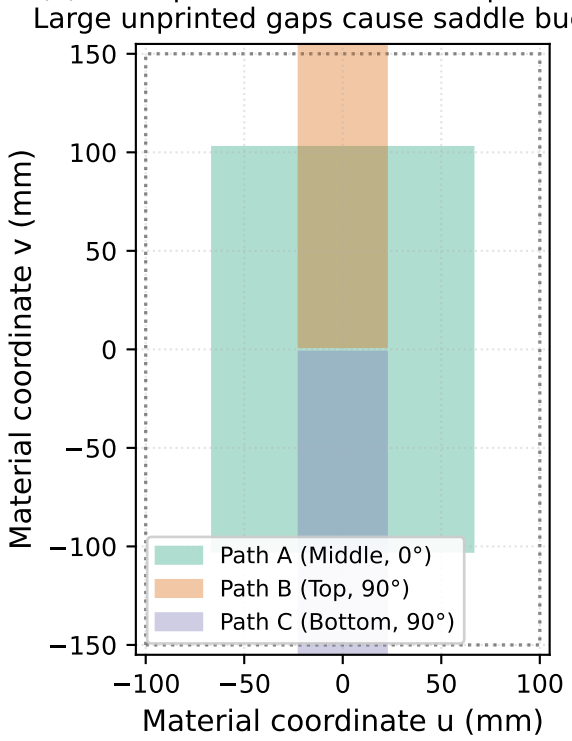


(d) Proposed Crosshatch: 18.2% Saddle Flux
Expansive continuous red synclastic dome core



Row 3: Toolpath Footprint & Layout on Sheet (200 x 300 mm)

(e) Attempt 045: Discrete 3-Toolpath Bands
Large unprinted gaps cause saddle buckling



(f) Proposed: Nested Pitch-Graded Terraces
Seamless orthogonal grading with zero gaps

